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Load matching for a current-steering digital-to-analog converter

Abstract

Certain aspects of the present disclosure are directed towards a digital-to-analog converter (DAC) system. The DAC system generally includes a first driver and a plurality of current-steering cells. A first current-steering cell of the plurality of current-steering cells includes: a first current source coupled to a first current-steering transistor and a second current-steering transistor, wherein a gate of the first current-steering transistor and a gate of the second current-steering transistor are coupled to a first output and a second output of the first driver, respectively; a first transistor having a source coupled to a current source path and a drain coupled to a reference potential node; and a second transistor having a source coupled to the current source path and a drain coupled to the reference potential node.

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Background/Summary

BACKGROUND

Field of the Disclosure

(1) Certain aspects of the present disclosure generally relate to electronic components and, more particularly, to circuitry for digital-to-analog conversion.

Description of Related Art

(2) Electronic devices include computing devices such as desktop computers, notebook computers,

tablet computers, smartphones, wearable devices like a smartwatch, internet servers, and so forth. These various electronic devices provide information, entertainment, social interaction, security, safety, productivity, transportation, manufacturing, and other services to human users. These various electronic devices depend on wireless communications for many of their functions. Wireless communication systems and devices are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such systems include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, and orthogonal frequency division multiple access (OFDMA) systems (e.g., a Long Term Evolution (LTE) system or a New Radio (NR) system). Wireless devices may include transmitters for processing signals for transmission via antennas. A transmitter may include one or more digital-to-analog converters (DACs) configured to convert signals from the digital domain to the analog domain for further processing (e.g., amplification) prior to transmission.

SUMMARY

(3) The systems, methods, and devices of the disclosure each have several aspects, no single one of which is solely responsible for its desirable attributes. Without limiting the scope of this disclosure as expressed by the claims which follow, some features will now be discussed briefly. After considering this discussion, and particularly after reading the section entitled “Detailed Description,” one will understand how the features of this disclosure provide the advantages described herein.

(4) Certain aspects of the present disclosure are directed towards a digital-to-analog converter (DAC) system. The DAC system generally includes a first driver and a plurality of current-steering cells. A first current-steering cell of the plurality of current-steering cells comprises: a first current source coupled to a first current-steering transistor and a second current-steering transistor, wherein a gate of the first current-steering transistor and a gate of the second current-steering transistor are coupled to a first output and a second output of the first driver, respectively; a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the first output of the first driver; and a second transistor having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the second output of the first driver.

(5) Certain aspects of the present disclosure are directed towards a method for digital-to-analog conversion. The method generally includes: receiving, at a first driver of a DAC, a first digital input via an input path of the DAC; and generating an analog output signal based on the first digital input by controlling, via the first driver, a first current-steering transistor and a second current-steering transistor of a first current-steering cell of a plurality of current-steering cells based on the first digital input, the first current-steering cell having a first current source coupled to the first current-steering transistor and the second current-steering transistor. The first current-steering cell further comprises: a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the first driver; and a second transistor having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the first driver.

(6) Certain aspects of the present disclosure are directed towards an apparatus for digital-to-analog conversion. The apparatus generally includes: means for receiving a first digital input; and a first current-steering cell of a plurality of current-steering cells coupled to the means for receiving, wherein the means for receiving includes means for controlling a first current-steering transistor and a second current-steering transistor of the first current-steering cell based on the first digital input, the first current-steering cell having a first current source coupled to the first current-steering transistor and the second current-steering transistor. The first current-steering cell further

comprises: a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the means for controlling; and a second transistor having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the means for controlling.

(7) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the appended drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects.

(2) FIG. 1 is a diagram of an example wireless communications network, in which aspects of the present disclosure may be practiced.

(3) FIG. 2 is a block diagram of an example access point (AP) and example user terminals, in which aspects of the present disclosure may be practiced.

(4) FIG. 3 is a block diagram of an example transceiver front end, in which aspects of the present disclosure may be practiced.

(5) FIG. 4 illustrates an example digital-to-analog converter (DAC), in accordance with certain aspects of the present disclosure.

(6) FIG. 5 and FIG. 6 illustrate example current-steering cells, in accordance with certain aspects of the present disclosure.

(7) FIG. 7 is a flow diagram depicting example operations for digital-to-analog conversion, in accordance with certain aspects of the present disclosure.

(8) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one aspect may be beneficially utilized on other aspects without specific recitation.

DETAILED DESCRIPTION

(9) Certain aspects of the present disclosure generally relate to techniques and apparatus for digital-to-analog conversion. For example, certain aspects provide a digital-to-analog converter (DAC) including current-steering cells implemented with an architecture that reduces timing skew errors across DAC segmentation boundaries. Timing skew errors degrade the noise floor and out-of-band DAC emissions. The timing skew errors may be reduced by connecting dummy transistors to switch driver outputs, as described in more detail herein.

(10) Example Wireless Communications

(11) FIG. 1 illustrates a wireless communications system **100** with access points **110** and user terminals **120**, in which aspects of the present disclosure may be practiced. For simplicity, only one access point **110** is shown in FIG. 1. An access point (AP) is generally a fixed station that communicates with the user terminals and may also be referred to as a base station (BS), an evolved Node B (eNB), a next generation Node B (gNB), or some other terminology. A user terminal (UT) may be fixed or mobile and may also be referred to as a mobile station (MS), an access terminal, user equipment (UE), a station (STA), a client, a wireless device, or some other terminology. A user terminal may be a wireless device, such as a cellular phone, a personal digital

assistant (PDA), a handheld device, a wireless modem, a laptop computer, a tablet, a personal computer, etc.

(12) Access point **110** may communicate with one or more user terminals **120** at any given moment on the downlink and uplink. The downlink (i.e., forward link) is the communication link from the access point to the user terminals, and the uplink (i.e., reverse link) is the communication link from the user terminals to the access point. A user terminal may also communicate peer-to-peer with another user terminal. A system controller **130** couples to and provides coordination and control for the access points.

(13) Wireless communications system **100** employs multiple transmit and multiple receive antennas for data transmission on the downlink and uplink. Access point **110** may be equipped with a number $N_{\text{sub.ap}}$ of antennas to achieve transmit diversity for downlink transmissions and/or receive diversity for uplink transmissions. A set $N_{\text{sub.u}}$ of selected user terminals **120** may receive downlink transmissions and transmit uplink transmissions. Each selected user terminal transmits user-specific data to and/or receives user-specific data from the access point. In general, each selected user terminal may be equipped with one or multiple antennas (i.e., $N_{\text{sub.ut}} \geq 1$). The $N_{\text{sub.u}}$ selected user terminals can have the same or different number of antennas.

(14) Wireless communications system **100** may be a time division duplex (TDD) system or a frequency division duplex (FDD) system. For a TDD system, the downlink and uplink share the same frequency band. For an FDD system, the downlink and uplink use different frequency bands. Wireless communications system **100** may also utilize a single carrier or multiple carriers for transmission. Each user terminal **120** may be equipped with a single antenna (e.g., to keep costs down) or multiple antennas (e.g., where the additional cost can be supported). In some aspects, the user terminal **120** or access point **110** may include a digital-to-analog converter (DAC) having current-steering cells implemented with an architecture for reducing timing skew errors, as described in more detail herein.

(15) FIG. 2 shows a block diagram of access point **110** and two user terminals **120_m** and **120_x** in the wireless communications system **100**. Access point **110** is equipped with $N_{\text{sub.ap}}$ antennas **224_a** through **224_{ap}**. User terminal **120_m** is equipped with $N_{\text{sub.ut,m}}$ antennas **252_{ma}** through **252_{mu}**, and user terminal **120_x** is equipped with $N_{\text{sub.ut,x}}$ antennas **252_{xa}** through **252_{xu}**. Access point **110** is a transmitting entity for the downlink and a receiving entity for the uplink. Each user terminal **120** is a transmitting entity for the uplink and a receiving entity for the downlink. As used herein, a “transmitting entity” is an independently operated apparatus or device capable of transmitting data via a frequency channel, and a “receiving entity” is an independently operated apparatus or device capable of receiving data via a frequency channel. In the following description, the subscript “dn” denotes the downlink, the subscript “up” denotes the uplink, $N_{\text{sub.up}}$ user terminals are selected for simultaneous transmission on the uplink, $N_{\text{sub.dn}}$ user terminals are selected for simultaneous transmission on the downlink, $N_{\text{sub.up}}$ may or may not be equal to $N_{\text{sub.dn}}$, and $N_{\text{sub.up}}$ and $N_{\text{sub.dn}}$ may be static values or can change for each scheduling interval. Beam-steering, beamforming, or some other spatial processing technique may be used at the access point and/or user terminal.

(16) On the uplink, at each user terminal **120** selected for uplink transmission, a transmitter (TX) data processor **288** receives traffic data from a data source **286** and control data from a controller **280**. TX data processor **288** processes (e.g., encodes, interleaves, and modulates) the traffic data {dup} for the user terminal based on the coding and modulation schemes associated with the rate selected for the user terminal and provides a data symbol stream {sup} for one of the $N_{\text{sub.ut,m}}$ antennas. A transceiver front end (TX/RX) **254** (also known as a radio frequency front end (RFFE)) receives and processes (e.g., converts to analog, amplifies, filters, and frequency upconverts) a respective symbol stream to generate an uplink signal. The transceiver front end **254** may also route the uplink signal to one of the $N_{\text{sub.ut,m}}$ antennas for transmit diversity via a radio frequency (RF) switch, for example. The controller **280** may control the routing within the transceiver front end

254. Memory **282** may store data and program codes for the user terminal **120** and may interface with the controller **280**.

(17) A number $N_{sub.up}$ of user terminals **120** may be scheduled for simultaneous transmission on the uplink. Each of these user terminals transmits its set of processed symbol streams on the uplink to the access point.

(18) At access point **110**, $N_{sub.ap}$ antennas **224a** through **224ap** receive the uplink signals from all $N_{sub.up}$ user terminals transmitting on the uplink. For receive diversity, a transceiver front end **222** may select signals received from one of the antennas **224** for processing. The signals received from multiple antennas **224** may be combined for enhanced receive diversity. The access point's transceiver front end **222** also performs processing complementary to that performed by the user terminal's transceiver front end **254** and provides a recovered uplink data symbol stream. The recovered uplink data symbol stream is an estimate of a data symbol stream $\{S_{up}\}$ transmitted by a user terminal. A receiver (RX) data processor **242** processes (e.g., demodulates, deinterleaves, and decodes) the recovered uplink data symbol stream in accordance with the rate used for that stream to obtain decoded data. The decoded data for each user terminal may be provided to a data sink (e.g., data sink **272m** or data sink **272x**, with corresponding data sink **244** for access point **110**) for storage and/or a controller **230** for further processing.

(19) On the downlink, at access point **110**, a TX data processor **210** receives traffic data from a data source **208** for $N_{sub.dn}$ user terminals scheduled for downlink transmission, control data from a controller **230** and possibly other data from a scheduler **234**. The various types of data may be sent on different transport channels. TX data processor **210** processes (e.g., encodes, interleaves, and modulates) the traffic data for each user terminal based on the rate selected for that user terminal. TX data processor **210** may provide a downlink data symbol streams for one or more of the $N_{sub.dn}$ user terminals to be transmitted from one of the $N_{sub.ap}$ antennas. The transceiver front end **222** receives and processes (e.g., converts to analog, amplifies, filters, and frequency upconverts) the symbol stream to generate a downlink signal. The transceiver front end **222** may also route the downlink signal to one or more of the $N_{sub.ap}$ antennas **224** for transmit diversity via an RF switch, for example. The controller **230** may control the routing within the transceiver front end **222**. Memory **232** may store data and program codes for the access point **110** and may interface with the controller **230**.

(20) At each user terminal **120**, $N_{sub.ut,m}$ antennas **252** receive the downlink signals from access point **110**. For receive diversity at the user terminal **120**, the transceiver front end **254** may select signals received from one or more of the antennas **252** for processing. The signals received from multiple antennas **252** may be combined for enhanced receive diversity. The user terminal's transceiver front end **254** also performs processing complementary to that performed by the access point's transceiver front end **222** and provides a recovered downlink data symbol stream. An RX data processor **270** processes (e.g., demodulates, deinterleaves, and decodes) the recovered downlink data symbol stream to obtain decoded data for the user terminal. In some aspects, the transceiver front end **254** or **222** may include a DAC having current-steering cells implemented with an architecture for reducing timing skew errors, as described in more detail herein.

(21) FIG. 3 is a block diagram of an example transceiver front end **300**, such as transceiver front ends **222**, **254** in FIG. 2, in which aspects of the present disclosure may be practiced. The transceiver front end **300** includes a transmit (TX) path **302** (also known as a "transmit chain") for transmitting signals via one or more antennas and a receive (RX) path **304** (also known as a "receive chain") for receiving signals via the antennas. When the TX path **302** and the RX path **304** share an antenna **303**, the paths may be connected with the antenna via an interface **306**, which may include any of various suitable radio frequency (RF) devices, such as a switch, a duplexer, a diplexer, a multiplexer, and the like.

(22) Receiving in-phase (I) or quadrature (Q) baseband analog signals from a digital-to-analog converter (DAC) **308**, the TX path **302** may include a baseband filter (BBF) **310**, a mixer **312**, a

driver amplifier (DA) **314**, and a power amplifier (PA) **316**. The BBF **310**, the mixer **312**, and the DA **314** may be included in a radio frequency integrated circuit (RFIC). In some cases, the PA **316** may be external to the RFIC.

(23) The BBF **310** filters the baseband signals received from the DAC **308**, and the mixer **312** mixes the filtered baseband signals with a transmit local oscillator (LO) signal to convert the baseband signal of interest to a different frequency (e.g., upconvert from baseband to RF). This frequency-conversion process produces the sum and difference frequencies of the LO frequency and the frequency of the signal of interest. The sum and difference frequencies are referred to as the “beat frequencies.” The beat frequencies are typically in the RF range, such that the signals output by the mixer **312** are typically RF signals, which may be amplified by the DA **314** and/or by the PA **316** before transmission by the antenna **303**. While one mixer **312** is illustrated, several mixers may be used to upconvert the filtered baseband signals to one or more intermediate frequencies and to thereafter upconvert the intermediate frequency (IF) signals to a frequency for transmission. In some aspects, the DAC **308** may include current-steering cells implemented with an architecture for reducing timing skew errors, as described in more detail herein.

(24) The RX path **304** includes a low noise amplifier (LNA) **322**, a mixer **324**, and a baseband filter (BBF) **326**. The LNA **322**, the mixer **324**, and the BBF **326** may be included in a radio frequency integrated circuit (RFIC), which may or may not be the same RFIC that includes the TX path components. RF signals received via the antenna **303** may be amplified by the LNA **322**, and the mixer **324** mixes the amplified RF signals with a receive local oscillator (LO) signal to convert the RF signal of interest to a different baseband frequency (i.e., downconvert). The baseband signals output by the mixer **324** may be filtered by the BBF **326** before being converted by an analog-to-digital converter (ADC) **328** to digital I or Q signals for digital signal processing.

(25) While it is desirable for the output of an LO to remain stable in frequency, tuning the LO to different frequencies typically entails using a variable-frequency oscillator, which may involve compromises between stability and tunability. Contemporary systems may employ frequency synthesizers with a voltage-controlled oscillator (VCO) to generate a stable, tunable LO with a particular tuning range. Thus, the transmit LO frequency may be produced by a TX frequency synthesizer **318**, which may be buffered or amplified by amplifier **320** before being mixed with the baseband signals in the mixer **312**. Similarly, the receive LO frequency may be produced by an RX frequency synthesizer **330**, which may be buffered or amplified by amplifier **332** before being mixed with the RF signals in the mixer **324**.

(26) While FIGS. 1-3 describe wireless communications as an example application in which certain aspects of the present disclosure may be implemented to facilitate understanding, certain aspects described herein may be used for digital-to-analog conversion in any of various other suitable applications (e.g., high-speed serializer/deserializer (SerDes) systems).

(27) Digital-to-Analog Conversion Using Current Steering

(28) Current-steering digital-to-analog converters (DACs) are the architecture of choice for high-performance digital-to-analog conversion in many wireless transmitters. Current-steering DACs offer versatility of design, allow high-speed operation, and provide high performance. In some cases, a segmented DAC may have different current-steering segments for processing of most-significant bits (MSBs) and least-significant bits (LSBs) of a digital input code. Some segmented DACs may suffer from timing errors across segmentation boundaries, creating short pulses in drive signals that provide wide bandwidth signal content causing increased noise floor and large out-of-band emission. Certain aspects provide a DAC architecture that reduces timing mismatch across DAC segmentation boundaries.

(29) FIG. 4 illustrates an example current-steering DAC **400** (e.g., corresponding to DAC **308** of FIG. 3), in accordance with certain aspects of the present disclosure. Depending on a digital input code, each bit of the DAC **400** associated with a current-steering cell may cause sourcing of a positive current or a negative current to respective outputs according to the logic level for a

respective bit of the digital input code. A “positive current” from a current-steering cell generally refers to a current provided to a positive output node (e.g., positive output node **426**), and a “negative current” from a current-steering cell generally refers to a current provided to a negative output node (e.g., negative output node **428**). Each of the current-steering cells may include switches (referred to herein as current-steering switches or transistors) to selectively provide a positive or a negative current to respective output nodes.

(30) As shown, the DAC **400** includes multiple current-steering cells, including most-significant bit (MSB) current-steering cells **402** used to process the MSBs of a digital input code of the DAC system, and least-significant bit (LSB) current-steering cells **404** used to process the LSBs of the digital input code. In some aspects, the DAC **400** may also include a middle (MID) current-steering cell (e.g., for a bit having a significance between the MSBs and LSBs). The LSB current-steering cells **404** may be coupled to a resistor-ladder circuit **440**. The resistor-ladder circuit **440** provides different weights associated with the LSB bits processed via the LSB current-steering cells **404**. As shown, drivers **408** may drive the current-steering transistors of the MSB current-steering cells **402**, driver **410** may drive the current-steering transistors of the current-steering cell **406**, and drivers **412** may drive the current-steering transistors of the current-steering cells **404**.

(31) As shown, each of the MSB current-steering cells **402** includes two current sources (e.g., p-channel metal-oxide-semiconductor (PMOS) transistors **414**, **416**) that are coupled to at least two positive current-steering transistors (e.g., PMOS transistors **418**, **420**, providing positive current to the positive output node **426**) and at least two negative current-steering transistors (e.g., PMOS transistors **422**, **424** providing negative current to the negative output node **428**). The sources of the PMOS transistors **418**, **420**, **422**, **424** may be coupled together and coupled to the drains of the PMOS transistors **414**, **416**. The PMOS transistors **418**, **420**, **422**, **424** may be referred to herein as current-steering transistors. The drains of the PMOS transistors **418**, **420** are coupled to the positive output node **426**, and the drains of the PMOS transistors **422**, **424** are coupled to the negative output node **428**, as shown.

(32) Each of the LSB current-steering cells **404** and the MID current-steering cell **406** may include fewer positive current-steering transistors and fewer negative current-steering transistors, as compared to each of the MSB current-steering cells **402**. For instance, the MID current-steering cell **406** may include a single positive current-steering transistor (e.g., PMOS transistor **430**) and a single negative current-steering transistor (e.g., PMOS transistor **432**), as opposed to the MSB current-steering cells **402** that each include two positive current-steering transistors and two negative current-steering transistors. The sources of PMOS transistors **430**, **432** may be coupled to a drain of PMOS transistor **434** used to implement a current source for the current-steering cell **406**. Each of the LSB current-steering cells **404** may be implemented similarly to the MID current-steering cell **406**. For instance, the LSB current-steering cell **450** may include a positive current-steering transistor (e.g., PMOS transistor **460**) and a negative current-steering transistor (e.g., PMOS transistor **462**). The sources of PMOS transistors **460**, **462** may be coupled to a drain of PMOS transistor **464** used to implement a current source for the current-steering cell **450**.

(33) As each of the MID and LSB current-steering cells includes fewer positive and negative current-steering transistors, the capacitive load for each of the drivers **408** may not be matched with the capacitive load for driver **410** or each of drivers **412**, causing a timing mismatch when driving the different current-steering cells. This timing mismatch causes timing skew errors across the DAC segmentation boundaries (e.g., a timing mismatch associated with digital outputs used to drive the current-steering cells **402** and current-steering cell **406**).

(34) In some implementations, switch drivers may be redesigned to have different output drive capacities to match the driver outputs' timings. However, this technique is challenging to track over process, voltage, and temperature (PVT) variations and causes harmonic distortion and increased noise floor. In some aspects of the present disclosure, non-conducting dummy transistors coupled to driver outputs may be used to match the capacitive loads of the drivers, as described in more

detail herein.

(35) FIG. 5 illustrates an example current-steering cell **500**, in accordance with certain aspects of the present disclosure. The current-steering cell **500** may correspond to the current-steering cell **406**, or any one of current-steering cells **404**, such as current-steering cell **450**. As shown, the current-steering cell **500** includes a PMOS transistor **506** (e.g., corresponding to PMOS transistor **430** of current-steering cell **406** or PMOS transistor **460** of current-steering cell **450**) and a PMOS transistor **508** (e.g., corresponding to PMOS transistor **432** of current-steering cell **406** or PMOS transistor **462** of current-steering cell **450**).

(36) The current-steering cell **500** may also include a PMOS transistor **514** which may be operated in saturation. A cascade current **516** (icas) may be provided to node **520** coupled to sources of transistors **506**, **508**, through PMOS transistor **514** (e.g., from source to drain of PMOS transistor **514**). Icas may be provided via the current source (e.g., PMOS transistor **434**) of the current-steering cell **406** or the current source (e.g., PMOS transistor **464**) of the current-steering cell **450**. As shown, a driver **502** (e.g., corresponding to one of drivers **412** or driver **410**) may be used to drive the gates of PMOS transistors **506**, **508**. For instance, a digital output (q) may be used to drive the gate of the PMOS transistor **506**, and a complementary digital output (qb) may be used to drive the gate of PMOS transistor **508**.

(37) In some aspects, in an attempt to match the load of driver **502** with the load of each of drivers **408**, the driver **502** may also drive the gate of a PMOS transistor **504** and the gate of a PMOS transistor **510**. The drains of PMOS transistors **504**, **510** are coupled to a reference potential node (e.g., electrical ground or vss, or any potential node that configures transistors **504**, **510** in saturation). The sources of PMOS transistors **504**, **510** are coupled to a drain of a PMOS transistor **512**. When the digital output (q) of the driver **502** is logic low, the PMOS transistor **504** is effectively configured as a diode-connected transistor (e.g., since the gate and the drain of the PMOS transistor **504** are at the same voltage potential), and when the complementary digital output (qb) of the driver **502** is logic low, the PMOS transistor **510** is effectively configured as a diode-connected transistor (e.g., since the gate and the drain of the PMOS transistor **510** are at the same voltage potential).

(38) The gates of PMOS transistors **512**, **514** are biased using a cascade voltage (vcas), as shown. The PMOS transistors **512**, **514** may be operated in saturation. A source current **518** (isource) may be provided to node **522** coupled to sources of PMOS transistors **504**, **510**. The source current **518** may also be referred to as bleeder current. As such, the current-steering cell **500** may be referred to as a current bleeder-based steering cell. Source current **518** may be provided through transistor **512** (e.g., from the source to the drain of transistor **512**). The source current **518** may be provided via a current source **540**. The PMOS transistor **512** reduces the impact of the line capacitance on the source side of the PMOS transistor **512** to the capacitance on the drain side of the PMOS transistor **512**.

(39) As shown, the current source **540** may also provide a source current to one or more other current-steering cells. The input impedance looking into the gates of the transistors **504**, **510** may be fairly constant over a wide range of bias currents (e.g., when the transistors **504**, **510** are in saturation). The current-steering cells may not be sensitive to small variations in the bias currents across the steering cells, allowing a single current source **540** to be used for providing bias currents for multiple current-steering cells, as described.

(40) With the PMOS transistors **504**, **510**, the load mismatch across the drivers may be reduced with a relatively small amount of increase in static power consumption for the DAC **400**. Moreover, the techniques described herein for reducing load mismatch provide an improved noise floor as compared to conventional implementations.

(41) FIG. 6 illustrates an example current-steering cell **600**, in accordance with certain aspects of the present disclosure. The current-steering cell **600** may correspond to any one of the MSB current-steering cells **402**. As shown, cascade current **616** (icas1) and cascade current **618** (icas2)

are provided via respective current sources (e.g., PMOS transistor **414** and PMOS transistor **416**). The current-steering cell **600** may include a PMOS transistor **612** and a PMOS transistor **614**. The cascade current **616** may be received at a source of the PMOS transistor **614** and the cascade current **618** may be received at a source of the PMOS transistor **612**. The drains of the PMOS transistor **612** and the PMOS transistor **614** may be coupled together and coupled to the sources of PMOS transistors **418**, **420**, **422**, **424**. The PMOS transistors **612**, **614** may be biased in saturation using a cascade voltage (vcas), as shown. The gates of the PMOS transistors **418**, **420**, **422**, **424** may be driven using a driver **602**. Driver **602** may correspond to one of the drivers **408**.

(42) FIG. 7 is a flow diagram depicting example operations **700** for digital-to-analog conversion, in accordance with certain aspects of the present disclosure. For example, the operations **700** may be performed by a DAC, such as the DAC **400**.

(43) The operations **700** begin, at block **702**, with the DAC receiving, at a first driver (e.g., driver **502** corresponding to driver **410** or driver **412**) of the DAC, a first digital input via an input path of the DAC. At block **704**, the DAC generates an analog output signal based on the first digital input by controlling, via the first driver, a first current-steering transistor (e.g., PMOS transistor **504**) and a second current-steering transistor (e.g., PMOS transistor **508**) of a first current-steering cell (e.g., current-steering cell **500**, corresponding to current-steering cell **406** or current-steering cell **450**) of a plurality of current-steering cells based on the first digital input. The first current-steering cell may have a first current source (e.g., PMOS transistor **434** or PMOS transistor **464**) coupled to the first current-steering transistor and the second current-steering transistor.

(44) In some aspects of the present disclosure, the first current-steering cell may also include a first transistor (e.g., PMOS transistor **504**) having a source coupled to a current source path (e.g., path providing source current **518**), a drain coupled to a reference potential node (e.g., vss), and a gate coupled to the first driver. The first current-steering cell may also include a second transistor (e.g., transistor **510**) having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the first driver.

(45) In some aspects, the DAC also receives, via a third transistor (e.g., transistor **512**) of the current source path, a first current (e.g., source current **518**) at a source of the third transistor, a drain of the third transistor being coupled to sources of the first transistor and the second transistor. The DAC may provide a second current via a second current source (e.g., current source **540**). The first current may include a first portion of the second current, a second portion of the second current being provided to a second current-steering cell of the plurality of current-steering cells. In some aspects, the first current-steering cell may also include a fourth transistor (e.g., transistor **514**) coupled between the first current source and a current-steering node (e.g., node **520**), the current-steering node being coupled to sources of the first current-steering transistor and the second current-steering transistor. A gate of the fourth transistor may be coupled to a gate of the third transistor.

(46) In some aspects, the DAC may also provide, via the first current-steering transistor, a first output current to a first output (e.g., positive output node **426**) of the DAC, a drain of the first current-steering transistor being coupled to the first output of the DAC. The DAC may provide, via the second current-steering transistor, a second output current to a second output (e.g., negative output node **428**) of the DAC, a drain of the second current-steering transistor being coupled to the second output of the DAC.

(47) In some aspects, the DAC may receive, at a second driver (e.g., driver **408**) of the DAC, a second digital input via the input path of the DAC. The DAC may generate the analog output signal based on the second digital input by controlling, via the second driver, a third current-steering transistor (e.g., PMOS transistor **420**) and a fourth current-steering transistor (e.g., PMOS transistor **422**) of a second current-steering cell (e.g., current-steering cell **402**) of the plurality of current-steering cells based on the second digital input. The second current-steering cell may include a second current source (e.g., PMOS transistor **414** or PMOS transistor **416**) coupled to the third

current-steering transistor and the fourth current-steering transistor. In some aspects, the second current-steering cell may include a fifth current-steering transistor (e.g., PMOS transistor **418**) having a source coupled to the second current source and a sixth current-steering transistor (e.g., PMOS transistor **424**) having a source coupled to the second current source. A gate of the fifth current-steering transistor and a gate of the sixth current-steering transistor may be coupled to a first output and a second output of the second driver, respectively. In some aspects, the first digital input (e.g., one of the LSBs or a bit processed by current-steering cell **406**) has a lower significance than the second digital input (e.g., MSB). A first load impedance for the first driver may be configured to match a second load impedance for the second driver. The first load impedance may be associated with the first current-steering cell and the second load impedance is associated with the second current-steering cell.

(48) In some aspects, drains of the first current-steering transistor and the second current-steering transistor may be coupled to a resistor-ladder circuit (e.g., resistor-ladder circuit **440**). The first current-steering transistor, the second current-steering transistor, the first transistor, and the second transistor may include PMOS transistors.

Example Aspects

(49) In addition to the various aspects described above, specific combinations of aspects are within the scope of the disclosure, some of which are detailed below:

(50) Aspect 1. A digital-to-analog converter (DAC) system, comprising: a first driver; and a plurality of current-steering cells, wherein a first current-steering cell of the plurality of current-steering cells comprises: a first current source coupled to a first current-steering transistor and a second current-steering transistor, wherein a gate of the first current-steering transistor and a gate of the second current-steering transistor are coupled to a first output and a second output of the first driver, respectively; a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the first output of the first driver; and a second transistor having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the second output of the first driver.

(51) Aspect 2. The DAC system of aspect 1, wherein the current source path comprises a third transistor configured to receive a first current at a source of the third transistor, a drain of the third transistor being coupled to the sources of the first transistor and the second transistor.

(52) Aspect 3. The DAC system of aspect 2, further comprising a second current source configured to provide a second current, wherein the first current comprises a first portion of the second current, a second portion of the second current being provided to a second current-steering cell of the plurality of current-steering cells.

(53) Aspect 4. The DAC system of any one of aspects 2-3, wherein the first current-steering cell further comprises a fourth transistor coupled between the first current source and a current-steering node, the current-steering node being coupled to sources of the first current-steering transistor and the second current-steering transistor.

(54) Aspect 5. The DAC system of aspect 4, wherein a gate of the fourth transistor is coupled to a gate of the third transistor.

(55) Aspect 6. The DAC system of any one of aspects 1-5, wherein: a drain of the first current-steering transistor is coupled to a first output of the DAC system; and a drain of the second current-steering transistor is coupled to a second output of the DAC system.

(56) Aspect 7. The DAC system of any one of aspects 1-6, wherein a second current-steering cell of the plurality of current-steering cells comprises: a second current source; a third current-steering transistor having a source coupled to the second current source; and a fourth current-steering transistor having a source coupled to the second current source, wherein a gate of the third current-steering transistor and a gate of the fourth current-steering transistor are coupled to a first output and a second output of a second driver, respectively.

(57) Aspect 8. The DAC system of aspect 7, wherein the second current-steering cell further

comprises: a fifth current-steering transistor having a source coupled to the second current source; and a sixth current-steering transistor having a source coupled to the second current source, wherein a gate of the fifth current-steering transistor and a gate of the sixth current-steering transistor are coupled to the first output and the second output of the second driver, respectively.

(58) Aspect 9. The DAC system of any one of aspects 7-8, wherein: the first current-steering cell is configured to process a first bit of a digital input code of the DAC system; and the second current-steering cell is configured to process a second bit of a digital input code of the DAC system, the first bit having a lower significance than the second bit.

(59) Aspect 10. The DAC system of any one of aspects 7-9, wherein: a first load impedance for the first driver is configured to match a second load impedance for the second driver; the first load impedance is associated with the first current-steering cell; and the second load impedance is associated with the second current-steering cell.

(60) Aspect 11. The DAC system of any one of aspects 1-10, wherein drains of the first current-steering transistor and the second current-steering transistor are coupled to a resistor-ladder circuit.

(61) Aspect 12. The DAC system of any one of aspects 1-11, wherein the first current-steering transistor, the second current-steering transistor, the first transistor, and the second transistor comprise p-channel metal-oxide-semiconductor (PMOS) transistors.

(62) Aspect 13. A method for digital-to-analog conversion, comprising: receiving, at a first driver of a digital-to-analog converter (DAC), a first digital input via an input path of the DAC; generating an analog output signal based on the first digital input by controlling, via the first driver, a first current-steering transistor and a second current-steering transistor of a first current-steering cell of a plurality of current-steering cells based on the first digital input, the first current-steering cell having a first current source coupled to the first current-steering transistor and the second current-steering transistor, wherein the first current-steering cell further comprises: a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the first driver; and a second transistor having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the first driver.

(63) Aspect 14. The method of aspect 13, further comprising receiving, via a third transistor of the current source path, a first current at a source of the third transistor, a drain of the third transistor being coupled to the sources of the first transistor and the second transistor.

(64) Aspect 15. The method of aspect 14, further comprising providing a second current via a second current source, wherein the first current comprises a first portion of the second current, a second portion of the second current being provided to a second current-steering cell of the plurality of current-steering cells.

(65) Aspect 16. The method of any one of aspects 14-15, wherein the first current-steering cell further comprises a fourth transistor coupled between the first current source and a current-steering node, the current-steering node being coupled to sources of the first current-steering transistor and the second current-steering transistor.

(66) Aspect 17. The method of aspect 16, wherein a gate of the fourth transistor is coupled to a gate of the third transistor.

(67) Aspect 18. The method of any one of aspects 13-17, further comprising: providing, via the first current-steering transistor, a first output current to a first output of the DAC, a drain of the first current-steering transistor being coupled to the first output of the DAC; and providing, via the second current-steering transistor, a second output current to a second output of the DAC, a drain of the second current-steering transistor being coupled to the second output of the DAC.

(68) Aspect 19. The method of any one of aspects 13-18, further comprising: receiving, at a second driver of the DAC, a second digital input via the input path of the DAC; and generating the analog output signal based on the second digital input by controlling, via the second driver, a third current-steering transistor and a fourth current-steering transistor of a second current-steering cell of the plurality of current-steering cells based on the second digital input, the second current-steering cell

having a second current source coupled to the third current-steering transistor and the fourth current-steering transistor.

(69) Aspect 20. The method of aspect 19, wherein the second current-steering cell comprises: a fifth current-steering transistor having a source coupled to the second current source; and a sixth current-steering transistor having a source coupled to the second current source, wherein a gate of the fifth current-steering transistor and a gate of the sixth current-steering transistor are coupled to a first output and a second output of the second driver, respectively.

(70) Aspect 21. The method of any one of aspects 19-20, wherein the first digital input has a lower significance than the second digital input.

(71) Aspect 22. The method of any one of aspects 19-21, wherein: a first load impedance for the first driver is configured to match a second load impedance for the second driver; the first load impedance is associated with the first current-steering cell; and the second load impedance is associated with the second current-steering cell.

(72) Aspect 23. The method of any one of aspects 13-22, wherein drains of the first current-steering transistor and the second current-steering transistor are coupled to a resistor-ladder circuit.

(73) Aspect 24. The method of any one of aspects 13-23, wherein the first current-steering transistor, the second current-steering transistor, the first transistor, and the second transistor comprise p-channel metal-oxide-semiconductor (PMOS) transistors.

(74) Aspect 25. An apparatus for digital-to-analog conversion, comprising: means for receiving a first digital input; and a first current-steering cell of a plurality of current-steering cells coupled to the means for receiving, wherein the means for receiving includes means for controlling a first current-steering transistor and a second current-steering transistor of the first current-steering cell based on the first digital input, the first current-steering cell having a first current source coupled to the first current-steering transistor and the second current-steering transistor, wherein the first current-steering cell further comprises: a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the means for controlling; and a second transistor having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the means for controlling.

ADDITIONAL CONSIDERATIONS

(75) Within the present disclosure, the word “exemplary” is used to mean “serving as an example, instance, or illustration.” Any implementation or aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects of the disclosure. Likewise, the term “aspects” does not require that all aspects of the disclosure include the discussed feature, advantage, or mode of operation. The term “coupled” is used herein to refer to the direct or indirect coupling between two objects. For example, if object A physically touches object B and object B touches object C, then objects A and C may still be considered coupled to one another—even if objects A and C do not directly physically touch each other. For instance, a first object may be coupled to a second object even though the first object is never directly physically in contact with the second object. The terms “circuit” and “circuitry” are used broadly and intended to include both hardware implementations of electrical devices and conductors that, when connected and configured, enable the performance of the functions described in the present disclosure, without limitation as to the type of electronic circuits.

(76) As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database, or another data structure), ascertaining, and the like. Also, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory), and the like. Also, “determining” may include resolving, selecting, choosing, establishing, and the like.

(77) The apparatus and methods described in the detailed description are illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes,

algorithms, etc. (collectively referred to as “elements”). The various operations or methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor. Generally, where there are operations illustrated in figures, those operations may have corresponding counterpart means-plus-function components with similar numbering. For example, means for receiving and means for controlling may include a driver, such as the driver **410** or drivers **412**.

(78) One or more of the components, steps, features, and/or functions illustrated herein may be rearranged and/or combined into a single component, step, feature, or function or embodied in several components, steps, or functions. Additional elements, components, steps, and/or functions may also be added without departing from features disclosed herein. The apparatus, devices, and/or components illustrated herein may be configured to perform one or more of the methods, features, or steps described herein.

(79) It is to be understood that the specific order or hierarchy of steps in the methods disclosed is an illustration of exemplary processes. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the methods may be rearranged. The accompanying method claims present elements of the various steps in a sample order, and are not meant to be limited to the specific order or hierarchy presented unless specifically recited therein.

(80) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. A phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover at least: a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c). All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

(81) It is to be understood that the claims are not limited to the precise configuration and components illustrated above. Various modifications, changes and variations may be made in the arrangement, operation and details of the methods and apparatus described above without departing from the scope of the claims.

Claims

1. A digital-to-analog converter (DAC) system, comprising: a first driver; and a plurality of current-steering cells, wherein a first current-steering cell of the plurality of current-steering cells comprises: a second current source configured to provide a second current, a first portion of the second current provided by a first current source and a second portion of the second current provided to a second current-steering cell of the plurality of current-steering cells; the first current source coupled to a first current-steering transistor and a second current-steering transistor, wherein

a gate of the first current-steering transistor and a gate of the second current-steering transistor are coupled to a first output and a second output of the first driver, respectively; a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the first output of the first driver; a second transistor having a source coupled to the current source path, a drain coupled to the reference potential node, and a gate coupled to the second output of the first driver; and a third transistor configured to receive a first current at a source of the third transistor, a drain of the third transistor being coupled to the sources of the first transistor and the second transistor.

2. The DAC system of claim 1, wherein the first current-steering cell further comprises a fourth transistor coupled between the first current source and a current-steering node, the current-steering node being coupled to sources of the first current-steering transistor and the second current-steering transistor.

3. The DAC system of claim 2, wherein a gate of the fourth transistor is coupled to a gate of the third transistor.

4. The DAC system of claim 1, wherein: a drain of the first current-steering transistor is coupled to a first output of the DAC system; and a drain of the second current-steering transistor is coupled to a second output of the DAC system.

5. The DAC system of claim 1, wherein the second current-steering cell of the plurality of current-steering cells comprises: a third current-steering transistor having a source coupled to the second current source; and a fourth current-steering transistor having a source coupled to the second current source, wherein a gate of the third current-steering transistor and a gate of the fourth current-steering transistor are coupled to a first output and a second output of a second driver, respectively.

6. The DAC system of claim 5, wherein the second current-steering cell further comprises: a fifth current-steering transistor having a source coupled to the second current source; and a sixth current-steering transistor having a source coupled to the second current source, wherein a gate of the fifth current-steering transistor and a gate of the sixth current-steering transistor are coupled to the first output and the second output of the second driver, respectively.

7. The DAC system of claim 5, wherein: the first current-steering cell is configured to process a first bit of a digital input code of the DAC system; and the second current-steering cell is configured to process a second bit of a digital input code of the DAC system, the first bit having a lower significance than the second bit.

8. The DAC system of claim 5, wherein: a first load impedance for the first driver is configured to match a second load impedance for the second driver; the first load impedance is associated with the first current-steering cell; and the second load impedance is associated with the second current-steering cell.

9. The DAC system of claim 1, wherein drains of the first current-steering transistor and the second current-steering transistor are coupled to a resistor-ladder circuit.

10. The DAC system of claim 1, wherein the first current-steering transistor, the second current-steering transistor, the first transistor, and the second transistor comprise p-channel metal-oxide-semiconductor (PMOS) transistors.

11. A method for digital-to-analog conversion, comprising: receiving, at a first driver of a digital-to-analog converter (DAC), a first digital input via an input path of the DAC; generating an analog output signal based on the first digital input by controlling, via the first driver, a first current-steering transistor and a second current-steering transistor of a first current-steering cell of a plurality of current-steering cells based on the first digital input, the first current-steering cell having a first current source coupled to the first current-steering transistor and the second current-steering transistor, wherein the first current-steering cell further comprises: a first transistor having a source coupled to a current source path, a drain coupled to a reference potential node, and a gate coupled to the first driver; and a second transistor having a source coupled to the current source

path, a drain coupled to the reference potential node, and a gate coupled to the first driver receiving, via a third transistor of the current source path, a first current at a source of the third transistor, a drain of the third transistor being coupled to the sources of the first transistor and the second transistor; and providing a second current via a second current source, wherein the first current comprises a first portion of the second current, a second portion of the second current being provided to a second current-steering cell of the plurality of current-steering cells.

12. The method of claim 11, wherein the first current-steering cell further comprises a fourth transistor coupled between the first current source and a current-steering node, the current-steering node being coupled to sources of the first current-steering transistor and the second current-steering transistor.

13. The method of claim 12, wherein a gate of the fourth transistor is coupled to a gate of the third transistor.

14. The method of claim 11, further comprising: providing, via the first current-steering transistor, a first output current to a first output of the DAC, a drain of the first current-steering transistor being coupled to the first output of the DAC; and providing, via the second current-steering transistor, a second output current to a second output of the DAC, a drain of the second current-steering transistor being coupled to the second output of the DAC.

15. The method of claim 11, further comprising: receiving, at a second driver of the DAC, a second digital input via the input path of the DAC; and generating the analog output signal based on the second digital input by controlling, via the second driver, a third current-steering transistor and a fourth current-steering transistor of a second current-steering cell of the plurality of current-steering cells based on the second digital input, the second current-steering cell having a second current source coupled to the third current-steering transistor and the fourth current-steering transistor.

16. The method of claim 15, wherein the second current-steering cell comprises: a fifth current-steering transistor having a source coupled to the second current source; and a sixth current-steering transistor having a source coupled to the second current source, wherein a gate of the fifth current-steering transistor and a gate of the sixth current-steering transistor are coupled to a first output and a second output of the second driver, respectively.

17. The method of claim 15, wherein the first digital input has a lower significance than the second digital input.

18. The method of claim 15, wherein: a first load impedance for the first driver is configured to match a second load impedance for the second driver; the first load impedance is associated with the first current-steering cell; and the second load impedance is associated with the second current-steering cell.

19. The method of claim 11, wherein drains of the first current-steering transistor and the second current-steering transistor are coupled to a resistor-ladder circuit.

20. The method of claim 11, wherein the first current-steering transistor, the second current-steering transistor, the first transistor, and the second transistor comprise p-channel metal-oxide-semiconductor (PMOS) transistors.
